Question Label : Multiple Select Question

Identify the set(s) of valid test cases for branch coverage.

Options :

6406534764267. ✖ {temperature = 0, temperature = 100}

6406534764268. ✔ {temperature = -5, temperature = 20, temperature = 50}

6406534764269. ✔ {temperature = -5, temperature = 20, temperature = 50, temperature = 105}

6406534764270. ✔ {temperature = -5, temperature = 20, temperature = 100, temperature = 50, temperature = 0}

Question Number : 45 Question Id : 6406531425407 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Identify the set(s) of valid test cases for multiple condition coverage.

Options :

6406534764271. ✖ {temperature = 0, temperature = 100}

6406534764272. ✖ {temperature = -5, temperature = 20, temperature = 50}

6406534764273. ✔ {temperature = -5, temperature = 20, temperature = 50, temperature = 105}

6406534764274. ✖ {temperature = -5, temperature = 20, temperature = 100, temperature = 50, temperature = 0}

Sw Testing

Section Id :	64065399899
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653230702
Question Shuffling Allowed :	No

Question Number : 46 Question Id : 6406531425408 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE TESTING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534764275. ✔ YES

6406534764276. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653230703
Question Shuffling Allowed :	Yes

Question Number : 47 Question Id : 6406531425409 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider that some modules are modified or added in an existing software program. Which of the following types of testing is done to ensure that all existing and new modules are working correctly?

Options :

6406534764277. ✖ Functional Testing

6406534764278. ✖ Performance Testing

6406534764279. ✔ Regression Testing

6406534764280. ✖ Usability Testing

Question Number : 48 Question Id : 6406531425410 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider that for testing a web application, the number of concurrent users was suddenly increased by a very large amount, and the performance of the application was measured. Which of the following types of testing does this scenario represent?

Options :

6406534764281. ✖ Stress Testing

6406534764282. ✔ Spike Testing

6406534764283. ✖ Soak Testing

6406534764284. ✖ Load Testing

Question Number : 49 Question Id : 6406531425411 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider a context free grammar G over a finite alphabet $\sum = \{a, b\}$. The following are the production rules for G:

$$S \rightarrow aXa$$
$$X \rightarrow aXb$$
$$X \rightarrow b$$

Let S be the starting symbol. Which of the following sets represents the language generated by the given grammar?

Options :

6406534764285. ✔ $\{a^n b^n a \mid n \geq 1\}$

6406534764286. ✖ $\{a^{n+1} b^n a \mid n \geq 1\}$

6406534764287. ✖ $\{a^n b^n a \mid n \geq 0\}$

6406534764288. ✖ $\{a^n b^{n+1} a \mid n \geq 0\}$

Question Number : 50 Question Id : 6406531425412 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Defect Removal Effectiveness (DRE) belongs to which of the following categories of software metrics?

Options :

6406534764289. ✖ Product Quality Metrics

6406534764290. ✖ Object-Oriented Software Metrics

6406534764291. ✖ Maintenance Quality Metrics

6406534764292. ✔ In-Process Quality Metrics

Question Number : 51 Question Id : 6406531425413 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Which type of client-side testing executes test cases that break the normal execution sequence of a web application?

Options :

6406534764293. ✖ Parameter level bypass testing

6406534764294. ✔ Control flow level bypass testing

6406534764295. ✖ Value level bypass testing

6406534764296. ✖ User-session data based testing

Question Number : 52 Question Id : 6406531425414 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

The following table lists the types of software artifacts and graph based models.

Artifact Type	Graph Model
1. Source Code	A. Finite State Machines
2. Design and Integration	B. Control Flow Graphs
3. Sequencing Constraints	C. Call Graphs

Match each type of software artifact with the appropriate graph based model that can used for testing it.

Options :

6406534764297. ✖ 1 - A, 2 - C, 3 - B

6406534764298. ✖ 1 - A, 2 - B, 3 - C

6406534764299. ✔ 1 - B, 2 - C, 3 - A

6406534764300. ✖ 1 - B, 2 - A, 3 - C

Question Number : 53 Question Id : 6406531425415 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider that a user fills a form in a web application, but instead of submitting the form, she clicks on the refresh button in the browser. Which type of transition would take place in the Application Transition Graph (ATG) in this scenario?

Options :

6406534764301. ✔ Operational Transition

6406534764302. ✖ Form Link Transition

6406534764303. ✖ Redirect Transition

6406534764304. ✖ Simple Link Transition

Question Number : 54 Question Id : 6406531425416 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following code segment.

```
1  class A {
2      protected String value;
3      public A() {
4          this.value = "basic";
5      }
6      public A(String value) {
7          this.value = value;
8      }
9      public void print() {
10         System.out.println(value);
11     }
12 }
13 class B extends A {
14     protected String value;
15     public B(String val) {
16         super.value = val;
17     }
18 }
```

A mutant needs to be created by replacing the `super` keyword in `LINE-16` with the `this` keyword. Which of the following mutation operators can be used for this purpose?

Options :

6406534764305. ✖ Hiding Variable Insertion (HVI)

6406534764306. ✔ Super Keyword Deletion (SKD)

6406534764307. ✖ Parent Constructor Deletion (PCD)

6406534764308. ✖ Overriding Method Deletion (OMD)

Question Number : 55 Question Id : 6406531425417 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider ISP with four blocks as $[A_1, A_2]$, $[B_1, B_2, B_3, B_4]$, $[C_1, C_2, C_3]$, and $[X_1, X_2]$. How many tests are needed to satisfy All Combinations Coverage (ACoC) criterion?

Options :

6406534764309. ✖ 12

6406534764310. ✖ 24

6406534764311. ✔ 48

6406534764312. ✖ 64

Question Number : 56 Question Id : 6406531425418 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following Java program.

```
class A {
    protected int x, y;
    public A() { x = y = 0; }
    public void setX(int val) { x = val; }
    public void setY(int val) { y = val; }
    public void print() {
        System.out.println("x = " + x + ", y = " + y);
    }
}
class B extends A {
    protected int y;
    public void setY(int val) { y = val; }
}
class C extends B {
    public void setX(int val) { x = (val + 10); }
    public void setY(int val) { y = val * 3; }
}
class Main {
    public static void main(String[] args) {
        C obj = new C();
        obj.setX(8);
        obj.setY(10);
        obj.print();
    }
}
```

The program outputs `x = 18, y = 0`, instead of `x = 18, y = 30`.

It is because the inherited variable `A::y` is overridden in the descendant `B` and therefore hides the instance variable `y` in `A`.

Which of the following types of anomalies or faults are present in the given scenario?

Options :

6406534764313. ✖ State Definition Anomaly (SDA)

6406534764314. ✔ State Definition Inconsistency Anomaly (SDIA)

6406534764315. ✖ Inconsistent Type Use (ITU)

6406534764316. ✖ State Visibility Anomaly (SVA)

Question Number : 57 Question Id : 6406531425419 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following Java program.

```
class A {
    public void u() { }
    public void v() { }
    public void w() { }
}
class B extends A {
    public void v() { }
    public void x() { }
}
class C extends B {
    public void u() { }
    public void x() { }
}
public class Test {
    public static void main(String[] args) {
        C obj = new C();
        obj.u();
        obj.v();
        obj.w();
        obj.x();
    }
}
```

Which of the following sets of methods will be invoked in the given program?

Options :

6406534764317. ✔ {C :: u(), B :: v(), A :: w(), C :: x()}

6406534764318. ✖ {C :: u(), A :: v(), A :: w(), C :: x()}

6406534764319. ✖ {A :: u(), A :: v(), A :: w(), C :: x()}

6406534764320. ✖ {A :: u(), B :: v(), A :: w(), B :: x()}

Question Number : 58 Question Id : 6406531425420 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question

Consider the following code segment for symbolic execution.

```
boolean isOdd(int n) {
    return n % 2 != 0;
}
void check(int x) {
    int k = (x + 1) * (x - 1);
    if(!isOdd(x)) {
        if(k < 10) {
          return;
        }
        abort();  // LINE-1
    }
}
```

Assume that x_0 respresents the symbolic value for the variable `x` in the method `void check(int x)`. Which of the following path constraints would reach the `abort()` call at `LINE-1`?

Options :

6406534764321. ✖ $(x_0\%2 = 0) \wedge ((x_0 * x_0) - 1 < 10)$

6406534764322. ✖ $(x_0\%2 \neq 0) \wedge ((x_0 * x_0) - 1 \geq 10)$

6406534764323. ✔ $(x_0\%2 = 0) \wedge ((x_0 * x_0) - 1 \geq 10)$

6406534764324. ✖ $(x_0\%2 \neq 0) \wedge ((x_0 * x_0) - 1 < 10)$

Question Number : 59 Question Id : 6406531425421 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following code statement (ground string) and a mutant:

```
// Ground String
int z = (a + b)*(a - b);

// Mutant-1
int z = (a + b)*(a + b);
```

Which of the following types of mutation operators is applied to the ground string to obtain `Mutant-1`?

Options :

6406534764325. ✖ Scalar variable replacement

6406534764326. ✔ Arithmetic operator replacement

6406534764327. ✖ Absolute value insertion

6406534764328. ✖ Assignment operator replacement

Question Number : 60 Question Id : 6406531425423 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following code segment of a Java servlet. The atomic sections are marked as $P_1, P_2, P_3, \cdots, P_7$.

	/* product_costs stores the cost of all purchases made of a particular product in a store over a period. */ ArrayList<Integer> product_costs = null; response.setContentType("text/html"); PrintWriter out=response.getWriter();

P_1	out.println ("<HTML><HEAD><TITLE>Product</TITLE></HEAD><BODY>");
	String productId = request.getParameter("product_id");
	/*
	getCosts() takes the prouduct ID (productId) as input, runs a query in the
	database, and returns a ArrayList object containing the costs for all
	purchases made for this product.
	*/
	product_costs = getCosts(productId);
	int total_cost = 0;
	double avg_cost = 0.0;

	if(product_costs == null) {
P_2	out.println("Invalid Product ID ");
	} else {
	for (Integer c : product_costs) {
P_3	total_cost += c;
	}
P_4	avg_cost = (double)total_cost / product_costs.size();
	if(avg_cost >= 30000.0)
P_5	out.println("Premium Product ");
	else
P_6	out.println("Standard Product ");
	}
P_7	out.println ("</BODY></HTML>"); out.close();

Identify the component expression that corresponds to the given code segment.

Options :

6406534764333. ✖ $P_1 \cdot (P_2 \mid (P_3^* \cdot (P_4 \cdot P_5) \mid P_6)) \cdot P_7$

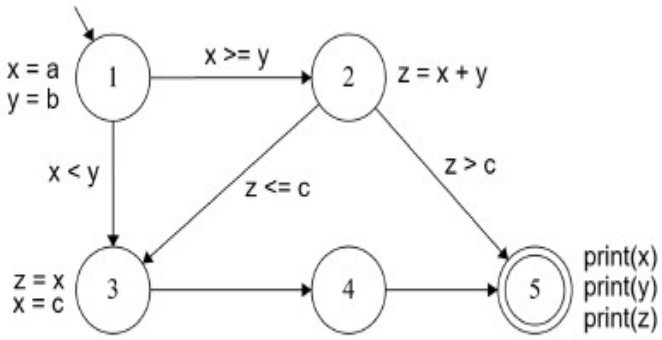
6406534764334. ✖ $P_1 \cdot ((P_2^* \cdot P_3 \cdot (P_4 \mid P_5)) \mid P_6) \cdot P_7$

6406534764335. ✖ $P_1 \cdot (P_2 \mid (P_3 \cdot P_4^* \cdot (P_5 \mid P_6))) \cdot P_7$

6406534764336. ✔ $P_1 \cdot (P_2 \mid (P_3^* \cdot P_4 \cdot (P_5 \mid P_6))) \cdot P_7$

Question Number : 61 Question Id : 6406531425424 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question

Consider the following data flow graph.



Given a set of test paths $\{[1,2,5],[1,3,4,5]\}$, which of the following coverage criteria is satisfied for the variable x?

Options :

6406534764337. ✖ All-Defs Coverage only

6406534764338. ✔ All-Uses Coverage only

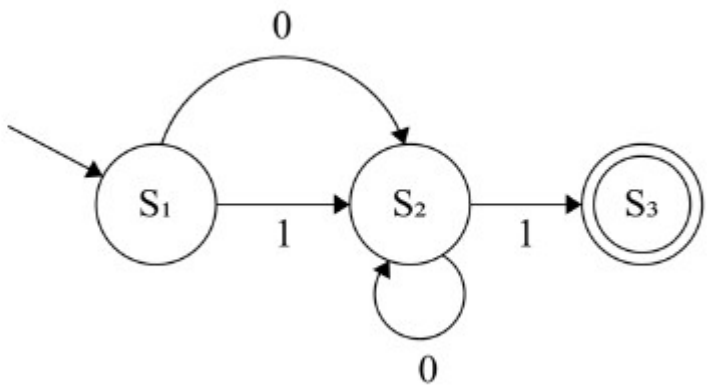
6406534764339. ✖ All-DU-Paths Coverage

6406534764340. ✖ None of these

Sub-Section Number :	3
Sub-Section Id :	640653230704
Question Shuffling Allowed :	Yes

Question Number : 62 Question Id : 6406531425422 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question

Consider the following finite state machine (FSM).



Which of the following strings can be accepted by the given FSM?

Options :

6406534764329. ✖ A string of only zeros

6406534764330. ✔ A string starting with one, followed by zeros, and ending with a one

6406534764331. ✖ A string starting with a one and ending with zeros

6406534764332. ✖ A string starting with a one, followed by zeros, and ending with a zero

Sub-Section Number :	4
Sub-Section Id :	640653230705
Question Shuffling Allowed :	Yes

Question Number : 63 Question Id : 6406531425425 Question Type : MSQ
Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider test-driven development (TDD) for a web portal for evaluating subjective assignments submitted by students. The teacher can evaluate an assignment and update the marks through this portal. A user story requires that the marks should be validated on submission. Consider the following tests for this user story.

```
public class TestMarksValidation {
    private MarksValidation obj;
    @Test
    public void test() {
        obj = new MarksValidation();
        assertEquals(obj.isValid(40), true);
        assertEquals(obj.isValid(30), false);
        assertEquals(obj.isValid(101), false);
        assertEquals(obj.isValid(99), true);
        assertEquals(obj.isValid(100), true);
    }
}
```

Which of the following are the appropriate implementations of the user story?

Options :

6406534764341. ✖

```java
public class MarksValidation {
    public boolean isValid(int marks) {
        if(marks <= 39) {
            System.out.println("Marks are less than the minimum");
            return false;
        }
        else { return true; }
    }
}
```

6406534764342. ✔

```java
public class MarksValidation {
    public boolean isValid(int marks) {
        if(marks < 40) {
            System.out.println("Marks are less than the minimum");
            return false;
        }
        else if(marks > 100) {
            System.out.println("Marks are more than the maximum");
            return false;
        }
        else { return true; }
    }
}
```

6406534764343. ✔

```java
public class MarksValidation {
    public boolean isValid(int marks) {
        if(marks <= 39) {
            System.out.println("Marks are less than the minimum");
            return false;
        }
        else if(marks >= 101) {
            System.out.println("Marks are more than the maximum");
            return false;
        }
        else { return true; }
    }
}
```

6406534764344. ✖

```java
public class MarksValidation {
    public boolean isValid(int marks) {
        if(marks <= 40) {
            System.out.println("Marks are less than the minimum");
            return false;
        }
        else if(marks >= 100) {
            System.out.println("Marks are more than the maximum");
            return false;
        }
        else { return true; }
    }
}
```

Sub-Section Number : 5
Sub-Section Id : 640653230706
Question Shuffling Allowed : No

Question Id : 6406531425426 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (64 to 65)

Question Label : Comprehension

Consider the predicate $p = (a \vee \neg b) \wedge \neg c$ and its truth table given below.
Note that the blank cells represent the value F in the truth table.

#	a	b	c	p	p_a	p_b	p_c
1	T	T	T				T
2	T	T		T	T		T
3	T		T				T
4	T			T			T
5		T	T				T
6		T			T	T	
7			T				T
8				T		T	

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 64 Question Id : 6406531425427 Question Type : MCQ Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following is the set of RACC pairs for clause *a*?

Options :

6406534764345. ✖ {(1, 5), (2, 6), (3, 7)}

6406534764346. ✔ {(2, 6)}

6406534764347. ✖ {(1, 5), (3, 7), (4, 8)}

6406534764348. ✖ {(1, 6)}

Question Number : 65 Question Id : 6406531425428 Question Type : MCQ Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following is the set of RICC pairs for clause *a*?

Options :

6406534764349. ✖ $\{(1,5),(2,6),(3,7)\}$ for $p = T$, no feasible pairs for $p = F$

6406534764350. ✖ $\{(2,6)\}$ for $p = T$, and $\{(1,6)\}$ for $p = F$

6406534764351. ✔ $\{(1,5),(3,7)\}$ for $p = F$, and $\{(4,8)\}$ for $p = T$

6406534764352. ✖ $\{(1,5),(3,7)\}$ for $p = T$, and $\{(4,8)\}$ for $p = F$

Question Id : 6406531425429 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (66 to 67)

Question Label : Comprehension

Consider the control flow graph (CFG), $G = \{V, E\}$, where

- Set of vertices $V = \{1,2,3,4,5,6,7\}$
- Set of edges $E = \{(1,2),(1,3),(2,4),(3,4),(4,5),(4,7),(5,6),(5,7)\}$
- Initial vertex $V_0 = 1$
- Final vertex $V_f = 7$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 66 Question Id : 6406531425430 Question Type : MCQ Correct Marks : 4

Question Label : Multiple Choice Question

What is the cyclomatic complexity of *G*?

Options :

6406534764353. ✖ 6

6406534764354. ✖ 5

6406534764355. ✖ 4

6406534764356. ✔ 3

Question Number : 67 Question Id : 6406531425431 Question Type : MCQ Correct Marks : 4

Question Label : Multiple Choice Question

How many test requirements are there for edge-pair coverage on *G*?

Options :

6406534764357. ✖ 6

6406534764358. ✖ 7